🛡️ PhishGuard

AI Email Phishing Detector with **Explainable Verdicts**

FutureTech HackFest 2026 · Applied AI × Cybersecurity · Chiara De Venuto

offline-firstprivacy-preserving98.9% accuracy

1 · The Problem

- Phishing is the #1 initial-access vector in cyberattacks — millions of new phishing sites per quarter.
- The real question is not "is this link blacklisted?" but **"is this email trustworthy?"** — before the click.
- Most detectors are **black boxes**: verdict without reasons → users never learn to defend themselves.
- Most detectors are **cloud-dependent**: email content leaves the device.

2 · Our Solution

PhishGuard — paste an email, get:

- 🎯 **Risk score 0-100** + verdict: SAFE / SUSPICIOUS / PHISHING
- 🧾 **Evidence**: active heuristic signals + top text indicators (the *why*)
- 🔒 **100% offline**: no API keys, no cloud, no data exfiltration
- ⚡ Web UI + JSON API

3 · How It Works

```
email text ──► TF-IDF (1-2 grams, 1677 features) ─┐
                         ├─► hstack ─► Logistic Regression ─► score 0-100
```

```
                    └─► 15 heuristic features (scaled) ─┘          │
                                                            verdict +
evidence ◄┘
```

- **ML layer** — captures subtle textual patterns.
- **Heuristic layer** — expert knowledge: typosquatting TLDs, raw-IP URLs, Reply-To mismatch, urgency keywords, attachments…
- **Interpretability by construction** — logistic coefficients are readable.

4 · Data & Methodology

- **720 emails**, balanced (360 phishing / 360 legit), generated deterministically (seed 42) from documented phishing patterns — **fully reproducible, code-committed**.
- **Hard examples** included: legit emails with phishing-like words ("verify your email", "password changed"), low-intensity phishing.
- Features: TF-IDF (1-2 grams) + **15 interpretable heuristics**.
- Model: **Logistic Regression** (class-balanced) — strong baseline, fast, explainable.
- Evaluation: stratified holdout + **5-fold CV** + ablation study.

5 · Results

Metric	Holdout (20%)	5-fold CV
Accuracy	**98.61%**	**98.75%**
Precision	**100%**	**100%**
Recall	**97.22%**	**97.50%**

F1	**98.59%**	**98.73%**

Holdout confusion matrix: [[72, 0], [2, 70]] — 0 false positives, 2 false negatives.

6 · Demo Flow

1️⃣ Paste any email (suspicious or normal)

2️⃣ Get the **score bar** + verdict badge

3️⃣ Read the **evidence**: active features & top indicators

4️⃣ Act on the advice — before clicking anything

100 **PHISHING** → 5 **SAFE**

Real output from the running app (see demo video).

7 · Innovation

- 🧠 **Explainability as a feature**: users learn *what to look for* — security awareness, the first line of defense.
- 🔒 **Offline-first, privacy-preserving**: email content never leaves the device.
- 🔬 **Reproducible corpus**: auditable, code-generated dataset.
- 🛡️ **Dual-layer defense**: ML + expert heuristics, both exposed.

8 · Limits & Roadmap

- Synthetic corpus → validate on real-world data (Enron + live phishing feeds).
- Single-token indicators → natural-language explanations.
- Logistic baseline → optional fine-tuned transformer (size vs. interpretability trade-off).

Roadmap

- Browser extension & email-client plugin
- Threat-intel blacklist as a third layer
- Explainability for the SUSPICIOUS band

9 · Stack

Python 3.12Flask 3.1scikit-learn 1.9TF-IDFLogistic Regressionjoblibunittest

Open-source only · No external services · MIT

GitHub: public repository with full code, dataset generator, tests and docs.

10 · AI Use Disclosure

This project was developed **with the assistance of generative AI tools** (AI-assisted coding, debugging and documentation). All code was executed, tested and verified locally; every metric comes from running the code.

Declared in compliance with:

- **Regulation (EU) 2024/1689 (EU AI Act), Article 50** — transparency obligations, applicable since 2 August 2026

- **Italian Law No. 132/2025** on artificial intelligence (Article 13)

Full transparency towards evaluators, users and third parties.

Thank you

🛡️ **PhishGuard** — Know why. Click safely.

FutureTech HackFest 2026 · Chiara De Venuto